

↗ What is a compiler

PDF (শ্রবণ পাঠ্য)

Th:

Terminology:

↗ Analysis - Synthesis model of Compilation

Two parts

↳ Analysis

↳ Synthesis

↗ "Symbol Table"

↗ Compilation Phases

Three Address Code = Abstract Code

Target Code = Machine Specific Instructions

❏ Functionalities
of Lexical Analyzer

- ① Stream of chars → Stream of tokens
- ② Lexeme → The characters comprising a token
Pattern →
- ③ Remove whitespaces & comments

* Phases were explained yet again from the PDF

x	+	=	y
1	2	3	4

x	=	x	+	y
1	2	3	4	5

Symbol Table

- ① Records the identifiers used in the source program
- ②

Semantic Error

myArray[5];

RealIndex = 10;

⇒ myArray[RealIndex];

// logically flawed

Syntax Analyzer & Lexical Analyzer



Recursive



Non-Recursive

Regeneration of
parent & child
nodes

Parent & child
node creations
only once

① Lexeme: A sequence of characters in the source code that are matched by given predefined patterns and thereby forms a valid token.

② Patterns: $\rightarrow$ (need context)

③ Patterns: A set of rules that a scanner follows to create a token.

$\rightarrow$ (regular expressions)

③ Token: The abstract category or name assigned to a group of lexemes that follow the same pattern.

$\rightarrow$ (chunk of text)

X Automata

int x = 10; 'Category categorization'

Token	Lexeme	Pattern
tok-int	int	int
tok-identifier	x	- / letter followed by - / letter / digits
tok-relation	=	=, <=, >=, ==, !=, <, >
tok-integer-constant	10	digit followed by 1/more digits
tok-symbol	;	;, :

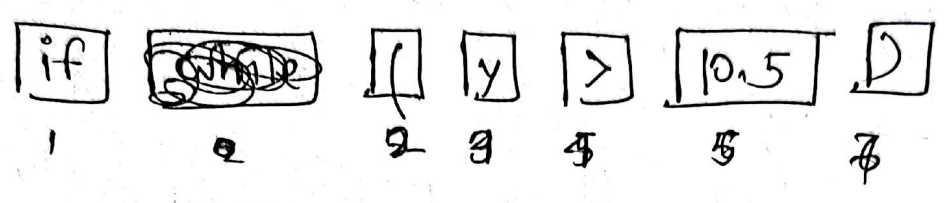
- ① while (x == 10)
 ② if (y > 10.5)

① [while] [(] [x] [==] [10] [)]
 1 2 3 4 5 6

Token	Lexeme	Pattern
tok-keyword	while	while
tok-parenthesis	(	(,), [,], {, }
tok-identifier	x	- / letter followed by - / letter / digits
tok-relation	==	==, <=, >=, <, > !=
tok-integer-constant	10	digit followed by 0 or more digits
tok-parenthesis	)	(,), [,], {, }

② if (y > 10.5)

r, →, v



Token	Lexeme	Pattern
tok-if	white if	if
tok-parenthesis	(	(,), [,], {, }
tok-identifier	y	-/letter followed by -/letter/digits
tok-relation	>	=, ==, <=, >=, <, >, !=
tok-float-constant	10.5	digit/digits followed by a 'dot' then 1 or more digits
tok-parenthesis	)	(,), [,], {, }

↗ Token Stream

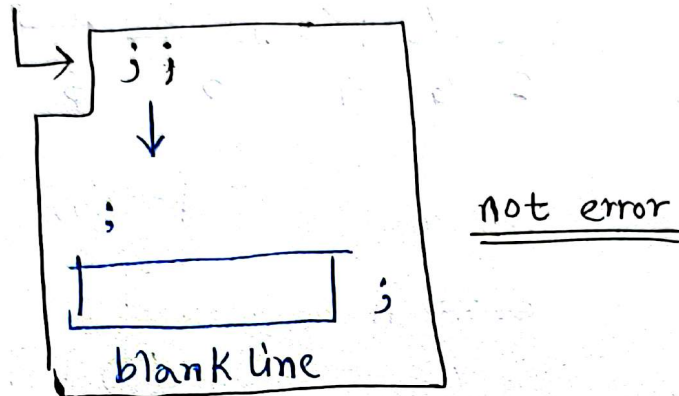
↗ Buffer Pair

↗ Lexical Error

PDF

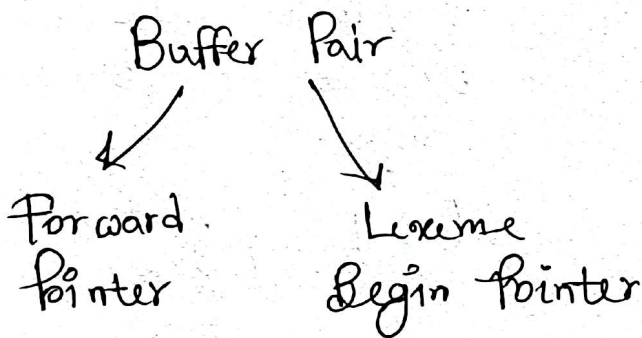
Error - Recovery Options :

① Deleting extra character

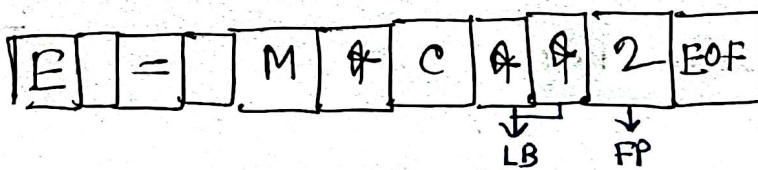


Managing Input Buffer

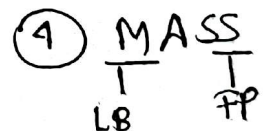
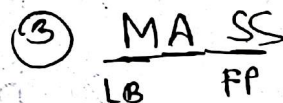
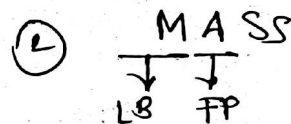
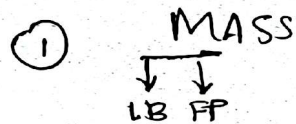
Temporary Storage



$$E = M * C * * 2$$



Lexeme Begin pointer = LB
Forward pointer = FP
They stay together



Sentinels

Null characters

Terminology
Grammar
next class